

Numerical Analysis First-Year Exam, August 2020

Do 4 (four) problems.

1. Let $\{\phi_k\}_{k=0}^{n+1}$ be a set of orthogonal polynomials on $[a, b]$, with respect to the inner product $(f, g) = \int_a^b f(x)g(x)w(x) dx$, indexed so that ϕ_k is of degree k . Prove that ϕ_k has k distinct roots $\{x_j^{(k)}\}_{j=1}^k$, with $x_j^{(k)} \in [a, b]$, $j = 1, \dots, k$.

2. Consider the interval $[a, b]$ with the partition $a = x_1 < x_2 < \dots < x_n < x_{n+1} = b$. Suppose $s(x)$ is the natural cubic spline that interpolates the data $\{(x_i, y_i)\}_{i=1}^{n+1}$, and that $g \in C^2[a, b]$ interpolates the same data. Show that

$$\int_a^b (s''(x))^2 dx \leq \int_a^b (g''(x))^2 dx.$$

3. This problem has two parts.

- (a) Consider the inner product on $C(0, 2)$ given by $(f, g) = \int_0^2 f(t)g(t) dt$. Find three orthonormal polynomials ϕ_0, ϕ_1, ϕ_2 on $(0, 2)$ with respect to the given inner product such that the degree of ϕ_n is equal to n , $n = 0, 1, 2$.
- (b) Find the nodes t_1 and t_2 and weights w_1 and w_2 which yield the weighted Gaussian Quadrature formula

$$\int_0^2 f(t) dt \approx w_1 f(t_1) + w_2 f(t_2)$$

with degree of exactness $m = 3$. You should find the nodes exactly, and may leave the weights w_1, w_2 in integral form.

4. Prove for any $f \in C[a, b]$ and integer $n \geq 0$, that the best uniform approximation of f in P_n is unique. You may assume the existence of at least one best uniform approximation of f .
5. Suppose $f \in C^{n+1}[a, b]$, and let $p \in P_n$ be a polynomial that interpolates the data $\{(x_i, f(x_i))\}_{i=0}^n$, where $x_0, \dots, x_n$ are distinct points in $[a, b]$. Consider an arbitrary fixed $x \in [a, b]$, and derive an exact expression for the error $f(x) - p(x)$.